⑥ $C_V = T\cdot\left(\frac{\partial S}{\partial T}\right)_V$ ; $TdS = C_p dT \Rightarrow C_p = T\cdot\left(\frac{\partial S}{\partial T}\right)_p$ (if $p=\text{const.}$)

$S = S(T;V)$: $dS = \left(\frac{\partial S}{\partial T}\right)_V dT + \left(\frac{\partial S}{\partial V}\right)_T dV \Rightarrow \left(\frac{\partial S}{\partial T}\right)_p = \left(\frac{\partial S}{\partial T}\right)_V + \left(\frac{\partial S}{\partial V}\right)_T \cdot \left(\frac{\partial V}{\partial T}\right)_p$

So:

$$C_p = \underbrace{T\cdot\left(\frac{\partial S}{\partial T}\right)_V}_{C_V} + \underbrace{T\cdot\left(\frac{\partial S}{\partial V}\right)_T}_{\left(\frac{\partial p}{\partial T}\right)_V}\cdot\underbrace{\left(\frac{\partial V}{\partial T}\right)_p}_{V\beta_p} = C_V + TV\beta_p\cdot\underbrace{\left(\frac{\partial p}{\partial T}\right)_V}_{\frac{\beta_p}{\kappa_T}} \Rightarrow$$

$\mathcal{F}$ (Maxwell)

previously: $\frac{\beta_p}{\kappa_T}$

$$C_p - C_V = \frac{TV\beta_p^2}{\kappa_T}$$

⑦ $\mu_{JL} = \left(\frac{\partial T}{\partial p}\right)_H$

$$\left(\frac{\partial T}{\partial p}\right)_H \cdot \left(\frac{\partial H}{\partial T}\right)_p \cdot \left(\frac{\partial p}{\partial H}\right)_T = -1$$

$$\left(\frac{\partial T}{\partial p}\right)_H = -\left(\frac{\partial H}{\partial p}\right)_T \cdot \frac{1}{\left(\frac{\partial H}{\partial T}\right)_p}$$

$H = U + pV \Rightarrow$

$dH = TdS + Vdp$

$p=\text{const.}$: $TdS = dH \Rightarrow$

$\Rightarrow C_p = \left(\frac{\partial H}{\partial T}\right)_p$

$dH = TdS + Vdp$

$$\left(\frac{\partial H}{\partial p}\right)_T = T\left(\frac{dS}{\partial p}\right)_T + V$$

Gibbs potential: $G(T;p) = U - TS + pV$

$dG = TdS - pdV - TdS - SdT + pdV + Vdp =$

$= -SdT + Vdp \Rightarrow$

$\Rightarrow \left(\frac{\partial G}{\partial T}\right)_p = -S$ ; $\left(\frac{\partial G}{\partial p}\right)_T = V \Rightarrow$

$\Rightarrow -\left(\frac{\partial S}{\partial p}\right)_T = \left(\frac{\partial V}{\partial T}\right)_p$

$$\left(\frac{\partial H}{\partial p}\right)_T = -T\cdot\left(\frac{\partial V}{\partial T}\right)_p + V \Rightarrow \mu_{JL} = \frac{1}{C_p}\cdot\left[T\left(\frac{\partial V}{\partial T}\right)_p - V\right]$$

ideal gas: $V = \frac{RT}{p} \Rightarrow T\left(\frac{\partial V}{\partial T}\right)_p - V = T \cdot \frac{R}{p} - V = 0$
$(n = 1\,\text{mol})$

for real gases: $\mu_{JK} > 0$ or $\mu_{JK} < 0$ → inversion temperature where $\mu_{JK}$ changes sign
gas cools down
→ liquefy gases

(8)

## Chemical potential

$dU = TdS - pdV + \delta W$
↳ change of the internal energy if a certain amount of particle gets into the system

$\delta W = \mu \cdot dN$
↓
intensive state variable

Gibbs potential: $G(T,p,N) = U - TS + pV = \mu N$

Euler's theorem:
$f(x_1, x_2, \dots) = \sum_i x_i \left(\frac{\partial f}{\partial x_i}\right)$ if $\lambda f(x_1, x_2, \dots) = f(\lambda x_1, \lambda x_2, \dots)$ (homogeneous function)

$\lambda G(T,p,N) = G(T,p,\lambda N) \Rightarrow G(T,p,N) = N \cdot \left(\frac{\partial G}{\partial N}\right)_{T,p} = \mu N$

$dG = -SdT + Vdp + \mu dN \Rightarrow \left(\frac{\partial G}{\partial N}\right)_{T,p} = \mu$

$T = \text{const.}: \left(\frac{\partial G}{\partial p}\right)_{T,N} = V \Rightarrow \left(\frac{\partial (\mu N)}{\partial p}\right)_{T,N} = V \Rightarrow$

$\Rightarrow N \cdot \left(\frac{\partial \mu}{\partial p}\right)_{T,N} = V \Rightarrow \mu = \int_{p_0}^{p} \frac{V}{N} \cdot dp + \mu_0 = k_B T \ln \frac{p}{p_0} + \mu_0$

$\frac{V}{N} = \frac{k_B T}{p}$